

Figure 6.3: Experimental PIV Window Visualization

at just under $y^+ = 5$. This undermines turbulence modeling without wall functions, but it should be noted that the value of y^+ only goes so high because the transitions to-and-from step changes in a pipe's cross-section were chosen to not have smooth axial grading as walls normally do in the system. This was done because for a fully-conformal and fully-hexahedral mesh, the wall refinement would be topologically required to propagate to the pipe's bulk flow region. This would result in a significant slow-down for the simulation's convergence on account of the pseudo-transient approach bounded by the Courant limit of 5 for minimal